ML Model Evaluation & Explainability Report

Generated on 2026-09-29 20:39:33 | Session ID: 8ba58df4-15f0-4a0c-835c-1e2185dff463

1. Dataset & Problem Detection Summary

Dataset File:	house_prices.csv	**Total Rows:**	2000
Target Column:	Sale_Price	**Total Columns:**	9
Problem Type:	Regression	**Test Split:**	50%

2. Preprocessing & Validation Protocol

Numerical Imputer	Median	**Categorical Encoding**	Onehot
Categorical Imputer	Most_frequent	**Feature Scaling**	Standard
Random State	42	**CV Protocol**	5 Folds

3. Model Performance Comparison

Rank	Model	MAE	MSE	RMSE	R^2	Adjusted R^2	Train Time (s)
1	Linear Regression	19765.12	601835236.09	24532.33	0.9823	0.9821	35.63
2	Ridge Regression	19779.90	601740170.27	24530.39	0.9823	0.9821	10.45
3	Random Forest Regressor	26648.97	1125737084.56	33552.01	0.9669	0.9665	4.57
4	Gradient Boosting Regressor	22849.07	807336419.99	28413.67	0.9763	0.9760	2.60
5	XGBoost Regressor	26484.93	1085627274.87	32948.86	0.9681	0.9677	2.78

4. Recommended Champion Model

Recommended Model: Linear Regression
Criterion: Highest R^2 (0.9823).
Rationale: Superior performance and fold consistency on held-out validation data.

5. Global Feature Importance (SHAP)

Feature	Mean Absolute SHAP Value	Relative Impact
Square_Feet	138570.1552	100.0%
Neighborhood_Suburbs	16018.8440	11.6%
Year_Built	15444.5667	11.1%
Bathrooms	12873.1010	9.3%
Condition_Rating	12128.5994	8.8%

Neighborhood_Uptown	11342.8499	8.2%
Garage_Cars	10612.4825	7.7%
Neighborhood_Downtown	10045.6584	7.2%